

# FUNDAMENTALS OF CROP AND ANIMAL SCIENCES

DRAFTED BY: ENGR. JOHN LUDOVIC  
ABELE 2024

## PRINCIPLES OF CROP SCIENCE

Crop science refers to the study of plant growth, management, and production of crops for food and other economic purposes.

### I. General Concepts

#### A. Plants vs. Crops

1. Plants - refers to any living organism under the kingdom Plantae.
2. Crops - refers to plants that are deliberately cultivated by humans for food and other purposes.

#### B. Classification of Crops

##### 1. Based on Growth Habit

- a. Herbs – succulent plants with self-supporting stems.
- b. Vines – succulent or woody plants without self-supporting stems.
- c. Shrubs – characterized by having several main branches arising from ground level with no trunk and typically growing not taller than 5 meters.
- d. Trees – characterized by having a single central trunk / stem to which branches are attached and are typically taller than shrubs.

##### 2. Based on Life Cycle

- a. Annual – refers to crops that complete their life cycle within one (1) year or less.
- b. Biennial – refers to crops that typically require two (2) growing seasons with a dormant period to complete their life cycles.
- c. Perennial – refers to crops that do not die after flowering and may produce seeds multiple times throughout their lifespan.

##### 3. Based on Reproduction

- a. Sexual – refers to crops that require the fertilization or the union of male and female gametes to produce seeds which will later be used to produce another crop.
- b. Asexual – refers to crops that may be propagated through vegetative means, producing genetically identical offsprings from the parent plant.

##### 4. Based on Cultivation Practice

- a. Agronomic – refers to crops that are typically grown in an extensive (large-scale) and often field based production, often considered as the foundation of the global food chain.
- b. Horticultural – refers to crops that are typically high-value which are grown more intensively (laborious and care-intensive), often on a smaller scale than agronomic crops.

##### 5. Based on Purpose

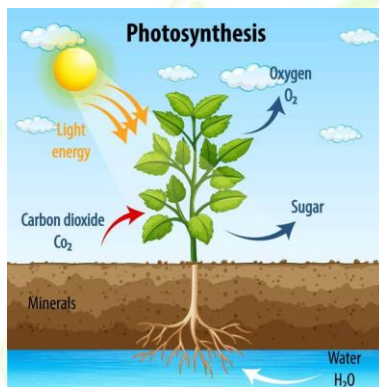
- a. Cereals and Grains – grown for their grains such as rice, corn, wheat, and sorghum.
- b. Legumes – grown for their pods and seeds such as stringbeans, cowpeas, mungbeans, and peanuts.
- c. Roots and tubers – grown for their enlarged roots / tuberous stems such as potatoes.
- d. Fiber crops – grown for their fibers such as cotton, ramie, kenaf, and jute.
- e. Oil crops – grown for their oil content such as coconut, castor, and sunflower.
- f. Sugar Crops – grown for their sugar content such as sugarcane and sugar beets.
- g. Forage Crops – grown as roughage source (animal feed) such as napier grass and ipil-ipil.

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- h. Beverage crops – grown for brewing beverages such as coffee, tea, and cacao.
  - i. Spices, condiments, and essences – grown for their special flavor, scent, or color such as vanilla, black pepper, and annatto.
  - j. Latex and resins – grown for their sap such as rubber tree.
  - k. Medicinal and Poison Crops – grown for their medicinal or poisonous properties such as lagundi, sambong, and tobacco.
  - l. Ornamentals – grown for their aesthetic value.
6. Special Classification of Crops
- a. Green Manure – crops that are grown, only to be plowed and incorporated into the soil to improve its fertility such as sesbania.
  - b. Cover Crop – grown to provide soil cover such as centrosema.
  - c. Companion Crop – crops that are grown together with another crop, benefitting either or both crops.
  - d. Trap Crop – crops that are planted to protect the main crop by attracting the pest to itself.
  - e. Catch Crop – a short-term crop planted between main crops to utilize the excess nutrients in the soil.
  - f. Soilage Crops – roughage that is cut and fed fresh to the animals.
  - g. Silage Crops – roughage that is cut and the fermented before being fed to the animals.

## II. Plant Life Processes



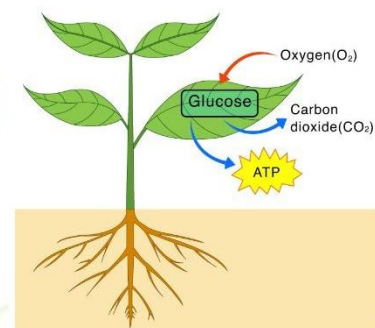
### Photosynthesis

This refers to the important process of “food making” in plants by utilizing the solar energy (or light energy) to produce sugars (chemical energy). Plants utilize carbon dioxide and water to produce glucose (stored energy of plants) and provides oxygen into the atmosphere.

This process takes place in the chloroplast. This organelle contains chlorophyll, the substance which gives the plants its green pigment. The main organ for photosynthesis is the leaf through the stomata which allows gas exchange for plants.

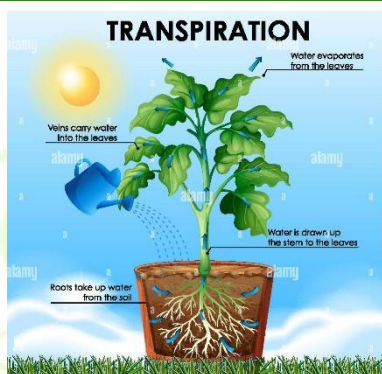
### Respiration

Just like humans, plants breathe. Respiration in plants refers to the process of producing energy for growth, reproduction, and other processes by breaking down stored energy (sugars) into immediate usable energy (Adenosine triphosphate or ATP).



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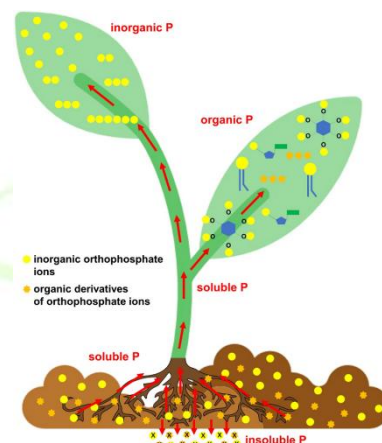


## Transpiration

The "necessarily evil" process of water loss in plants. This process is crucial for plant functions such as nutrient uptake, photosynthesis, and cooling processes. It also maintains favorable turgor pressure (pressure inside the cells) which permits the uptake of nutrients from the soil.

## Translocation

The process of transporting organic substances from where they are produced to where they are needed—basically, the logistics of plants. Assimilates are transported through the phloem from the leaves to the other parts of the plants.



## Assimilation

This refers to the process of converting absorbed nutrients and stored energy into complex organic molecules, primarily amino acids, which are essential for plant growth.

### a. Nitrogen Assimilation Process

1. Nitrogen Fixation
2. Nitrification
3. Ammonification
4. Denitrification

## III. Plant Growth Regulation

This process involves the interplay of different hormones and other factors to control the development of the plant. The following are the key plant hormones and their functions during in crop development.

|                              |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|
| <b>Auxins</b>                | Promotes stem elongation and root initiation during the vegetative growth of plants. |
| <b>Gibberellic Acid (GA)</b> | Promotes seed germination and fruit enlargement.                                     |
| <b>Cytokinin</b>             | Promotes shoot growth and nutrient mobilization.                                     |
| <b>Abscisic Acid (ABA)</b>   | Enhances stomatal functions.                                                         |

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|                 |                                                              |
|-----------------|--------------------------------------------------------------|
| <b>Ethylene</b> | The only gaseous phytohormone which promotes fruit ripening. |
|-----------------|--------------------------------------------------------------|

**Parthenocarp** - This refers to the process of producing fruits without fertilization, resulting in seedless fruits. This may occur naturally (like in bananas), through hormonal triggers (plants overproduce phytohormones), or induced (artificial introduction of auxins and GAs).

## IV. Factors Affecting Crop Production

The growth, development, and yield of a crop is a result of its genotype, its environment, and the interaction between the two.

1. Genotype refers to the genetic makeup of an organism. In crop science, the genotype of a crop provides information regarding the type of environment where it must be cultivated and dictates the ceiling of the yield that it may produce.
2. Environment refers to any external factor that influences the growth and development of the crop including climatic, edaphic, biotic, and human factors.
3. The interaction between the genotype and the environment may be described by the following statements:
  - A high yielding variety grown under unfavorable conditions will still produce a low yield; and
  - A low yielding variety grown in an optimum environment will still have a low yield.

### A. Climatic Factors

- a. Weather – refers to the momentous condition of the atmosphere at a certain geographic location.
  - b. Climate – refers to the seasonal pattern of a certain geographic location. This is an aggregate of the day-to-day weather conditions, usually presented as averages and variability.
    1. Macroclimate – refers to the overall climate of a large geographic region. This dictates what type of crops may be cultivated in the area.
    2. Microclimate – refers to the localized climate within a specific portion of a region, typically below the canopy of the crop. This influences how well a specific crop will perform if cultivated in the area.
1. **Precipitation** – this refers to water falling from the atmosphere in any form. This is an important factor as water serves a crucial purpose in crop growth and development. This serves as a source of moisture which the plant needs for its life processes.

Plants may also be classified based on their need for moisture:

- a. Xerophytes – plants that had adapted to survive in arid regions;
  - b. Hydrophytes – plants that had adapted to survive in aquatic or submerged environment; and
  - c. Mesophytes – adapted to moderate or well-watered environments.
2. **Temperature** – this refers to the hotness or coldness in the area. In agriculture, there are three cardinal temperatures:
    - a. Minimum Temperature –

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- b. Optimum Temperature –
- c. Maximum Temperature –

Plants may also be classified based on their temperature requirement:

- a. Warm-season crops
  - b. Cold-season crops –
  - c. Tropical crops –
  - d. Sub-tropical crops –
3. **Wind** – this refers to the movement of air and affects plants depending on its velocity. Strong winds cause destructive damages to plants, sterility due to loss of pollens, reduced photosynthetic activity due to loss of leaves and dispersal of carbon dioxide.
4. **Solar Radiation** – this refers to the amount of sunlight or energy given out by the sun. Plants react to sunlight based on intensity, duration, and wavelength. It also influences the overall temperature of the region.
- a. **Photoperiodic** – depending on the duration of sunlight.
  - b. **Photoenergetic** – depending on the wavelength of the radiation.
  - c. **Photocybernetic** – depending on the quality of solar radiation.

Plants may also be classified based on their need for sunlight:

- a. **Heliophytes** – sun-loving crops such as mango, banana, and coconut.
- b. **Sciophytes** – shade-loving crops such as black pepper and cacao.

Plants may also be classified according to photoperiodic response:

- a. **Day-neutral** – induces flowering over a wide range of daylength.
  - b. **Long-day plant** - inhibits flowering when exposed to extended periods without sunlight / solar radiation.
  - c. **Short-day plant** – requires an extended dark period to induce flowering.
5. **Relative Humidity** – this refers to the amount of water in the air relative to its capacity. This affects the growth and development of crops by influencing nutrient uptake, photosynthesis, and susceptibility to diseases.

## B. Edaphic Factors

The soil is a mixture of organic and inorganic materials developed on the Earth's surface through the weathering of rocks and minerals which serves as a medium of plant growth. In between the solid components of the soil are spaces which are filled with water and air which are utilized by plants for biochemical processes such as photosynthesis.

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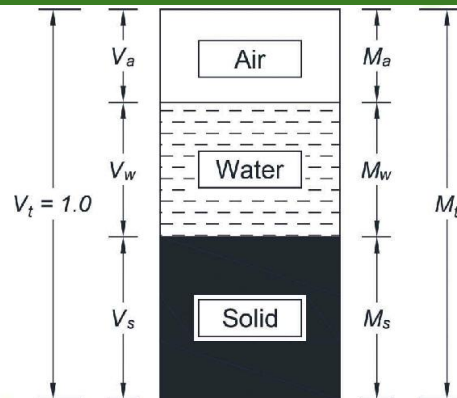


Figure 1. Composition of soil mass and volume.

The **mass (denoted as m)** of soil is composed of three components: solids (soil particles), water, and air; however, the mass of air is considered negligible and is assumed to be zero.

Equation 1.

$$\begin{aligned} m_{\text{total}} &= m_{\text{solid}} + m_{\text{water}} + m_{\text{air}} \\ m_{\text{total}} &= m_{\text{solid}} + m_{\text{water}} \end{aligned}$$

The **volume (denoted as V)** of soil is composed of three components: solids (soil particles), water, and air. In the case of soil volume, the volume of air is not neglected as the void spaces are not always saturated with moisture.

Equation 2.

$$V_{\text{total}} = V_{\text{solid}} + V_{\text{water}} + V_{\text{air}}$$

The volume occupied by water and air are called void spaces; hence, the volume of soil may also be characterized as.

Equation 3.

$$\begin{aligned} V_{\text{total}} &= V_{\text{solid}} + V_{\text{void}} \\ \text{Where } V_{\text{void}} &= V_{\text{water}} + V_{\text{air}} \end{aligned}$$

## A. Soil Properties Relevant to Crop production

1. **Soil Texture** – refers to the relative proportion of sand, silt, and clay particles in the soil. This influences the rate of infiltration, retention, and drainage of water in the soil—generally, coarser soils have higher infiltration rates than finer soils; however, they have lower water holding capacity compared to finer soils.

| Soil Particle | Particle Size      |
|---------------|--------------------|
| Coarse Sand   | 2.00 mm – 0.20 mm  |
| Fine Sand     | 0.20 mm – 0.02 mm  |
| Silt          | 0.02 mm – 0.002 mm |
| Clay          | Less than 0.002 mm |

### a. Classification of Soil based on Texture

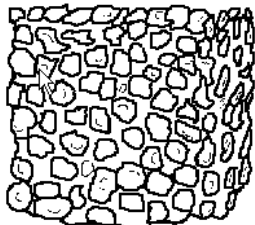
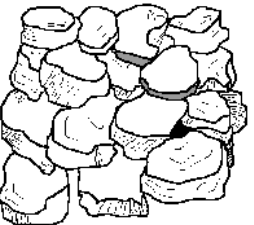
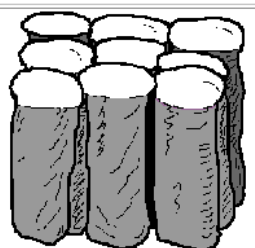
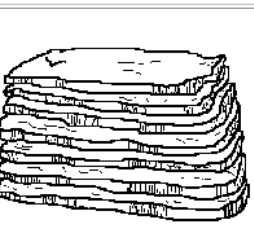
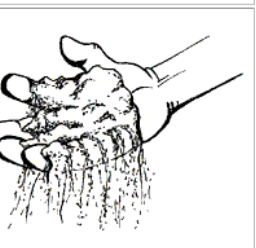
1. **Loam** – neither sand, silt, and clay particles are predominant.

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2. Sandy Loam – sand particles are slightly predominant.
3. Silt Loam – silt particles are slightly predominant
4. Clay Loam – clay particles are slightly predominant.

**2. Soil Structure** - refers to the clustering of soil particles into aggregates of various shapes, sizes, and stability. This influences the water infiltration and the aeration of the soil—for example, compacted soils with platy structures impede the infiltration of water into the soil.

|                                                                                                                                                         |                                                                                                 |                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        |               |                                                                                 |
| <b>Granular:</b> Resembles cookie crumbs and is usually less than 0.5 cm in diameter. Commonly found in surface horizons where roots have been growing. | <b>Blocky:</b> Irregular blocks that are usually 1.5 - 5.0 cm in diameter.                      | <b>Prismatic:</b> Vertical columns of soil that might be a number of cm long. Usually found in lower horizons.                                                     |
|                                                                       |              |                                                                                |
| <b>Columnar:</b> Vertical columns of soil that have a salt "cap" at the top. Found in soils of arid climates.                                           | <b>Platy:</b> Thin, flat plates of soil that lie horizontally. Usually found in compacted soil. | <b>Single Grained:</b> Soil is broken into individual particles that do not stick together. Always accompanies a loose consistence. Commonly found in sandy soils. |

**3. Soil Density** – denoted as  $\rho$ , this refers to the “extent of compaction” of the soil expressed in units of mass per volume. There following are the two measures of soil density, as follows:

- a. **Bulk Density** – refers to the ratio of the mass of the soil particles to its total undisturbed volume. The common value of soil bulk density is 1.25 g/cc.

$$\rho_{\text{bulk}} = m_{\text{solid}} / V_{\text{total}}$$

- b. **Particle Density** - refers to the ratio of the mass of the soil particles to its volume. The common value of soil particle density is 2.65 g/cc.

$$\rho_{\text{particle}} = m_{\text{solid}} / V_{\text{solid}}$$

**4. Soil Porosity** – denoted as  $n$ , this refers to the volume of the void spaces expressed as a percentage of the total undisturbed volume of the soil. This influences the air and water movement within the soil which is an essential factor for crop growth.

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$$n = V_{\text{void}} / V_{\text{total}} \\ = 1 - (\rho_{\text{bulk}} / \rho_{\text{particle}})$$

5. **Soil Specific Gravity** – denoted as SG, this refers to the ratio of the density of the soil to the density of water. There are also two measures of soil specific gravity, as follows:

a. **Apparent Specific Gravity** – refers to the ratio of the soil bulk density to the density of water.

$$ASG = \rho_{\text{bulk}} / \rho_{\text{water}}$$

b. **Real Specific Gravity** – refers to the ratio of the soil particle density to the density of water.

$$RSG = \rho_{\text{particle}} / \rho_{\text{water}}$$

6. **Soil Moisture Content** – this refers to the amount of water in the soil expressed as a percentage of its mass or volume. This may be calculated using the following bases:

a. **Wet Mass Basis** – refers to the percentage of soil moisture in reference to the mass of wet or moist soil.

$$SMC_{\text{wb}} = m_{\text{water}} / m_{\text{total}} \\ = (m_{\text{initial}} - m_{\text{final}}) / m_{\text{initial}}$$

b. **Dry Mass Basis** – refers to the percentage of soil moisture in reference to the mass of the soil after drying.

$$SMC_{\text{db}} = m_{\text{water}} / m_{\text{solids}} \\ = (m_{\text{initial}} - m_{\text{final}}) / m_{\text{final}}$$

c. **Volumetric Basis** – refers to the percentage of soil moisture in reference to the total undisturbed volume of the soil.

$$SMC_v = V_{\text{water}} / V_{\text{total}} \\ = SMC_{\text{db}} \times ASG$$

Soil moisture content may also be expressed in terms of depth the equation:

$$d_{\text{moisture}} = SMC_v \times d_{\text{soil}} \\ = SMC_{\text{db}} \times ASG \times d_{\text{soil}}$$

## Types of Soil Moisture

1. **Gravitational Water** – refers to soil moisture that is easily drained by the force of gravity.
2. **Capillary Water** – refers to soil moisture is bound to the soil by capillary forces (cohesive forces) at a tension of 1/10 to 1/3 bar.
3. **Hygroscopic Water** – refers to soil moisture that is tightly bound to the soil particles by adhesive forces at a tension of 15 bars and can only be removed by oven drying at 105 °C.

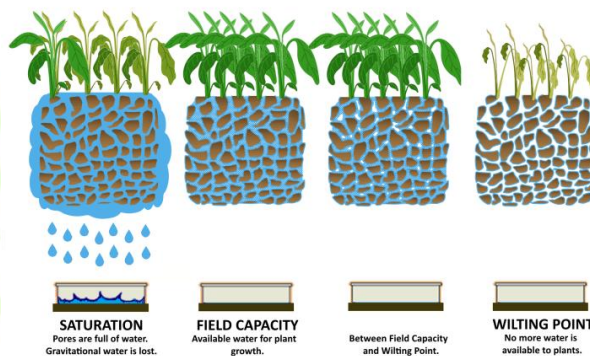
## Soil Moisture Constants

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1. **Field Capacity (FC)** - this refers to the amount of water that the soil can hold against the force of gravity usually one to two days after irrigation. This is the ceiling of the amount of water that is available for plants.
2. **Permanent Wilting Point (PWP)** - this refers to the amount of water which is can no longer be utilized by plants, and they will start to permanently wilt.
3. **Saturation Point** - this refers to when all the void spaces in the soil are filled with water. A saturated soil is generally detrimental to plants as it limits air in the root zone.
4. **Water Holding Capacity (WHC) / Available Moisture (AM)** - the difference between field capacity and permanent wilting point, which refers to the amount of moisture that is available for plant utilization.

$$WHC = FC - PWP$$



Soil moisture constants.

7. **Soil pH** – this refers to the acidity or alkalinity of the soil. pH influences the availability of nutrients in the soil—extreme acidity or alkalinity of the soil leads to stunted growth of crops.
8. **Soil Cation Exchange Capacity** – this refers to the ability of the soil to absorb and exchange cations for nutrient uptake.
9. **Soil Organic Matter** – this refers to the number of carbon-containing compounds in the soil. Organic matter enhances soil aggregation, water infiltration and retention. It also helps in managing CEC and the nutrient availability in the soil.

## V. Farming and Cropping Systems

### A. General Types of Farming Systems

1. Lowland Crop Production –
2. Upland Crop Production
3. Dryland Crop Production
4. Agro-Forestry

### B. Cropping Systems

This refers to the arrangement of crops in time and space as well as the process of growing them. The following are the common systems of crop production:

1. **Monocropping or Monoculture** – this refers to the cultivation of only one type of crop in a parcel of land within a year.
2. **Crop Rotation** – this refers to the cultivation of more than one crop in the same parcel of land, in sequence, within a year.

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3. Intercropping – multiple crops are grown simultaneously on the same field such that the period of overlap is long enough to include the vegetative stage.
4. Relay Cropping - multiple crops are grown together but the secondary crop is planted only after the flowering stage of the primary crop.
5. Ratoon Cropping - a practice where a crop is harvested multiple times from the same planting.
6. Sloping Agricultural Land Technology - crops are planted on sloping lands also as an engineering measure for soil erosion.

## C. Field Patterns / Field Layouts

- a. Row Cropping
- b. Strip Cropping
- c. Alley Cropping
- d. Orchard Layout
  1. Square Layout
  2. Triangle Layout
  3. Quincunx Layout

## VI. Land Preparation

Land preparation is an important process to ensure that favorable soil conditions are provided for the germination and growth of a particular crop. It is crop-dependent—land should be tilled according to what crop will be grown and what system will be used.

### A. Tillage

Tillage refers to the mechanical manipulation of soil to provide the proper medium for the establishment and growth of crops. The following are the different methods and systems of soil tillage.

1. Types of Tillage Systems
  - a. Conservation Tillage – a system that maintains at least 30% residue cover on the soil surface after planting.
  - b. Conventional Tillage – a system that is traditionally performed in preparing the seed bed for a specific crop in a specific geographic location.
  - c. Minimum Tillage – minimal soil manipulation is performed
  - d. Mulch Tillage – a system comparable to conservation tillage, in which plant residue is left on or near the soil surface for moisture retention and erosion control.
  - e. Optimum Tillage - an idealized system which permits the maximum net return for a crop under given conditions.
  - f. Precision Tillage – a system where primary tillage is done in conjunction with special planting procedures to reduce secondary tillage operation.
  - g. Reservoir Tillage - system where large number of depressions are deliberately left on the field to hold water.
  - h. Ridge Tillage - a system where ridges are formed during cultivation.
  - i. Strip Tillage - only isolated bands of land are tilled.
2. Types of Tillage Operations
  - a. Primary Tillage - this refers to the initial soil working operation to reduce soil strength, cover plant materials, and rearrange aggregates, at depths of <insert depth>. This is done using plows and subsoilers.

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- b. Secondary Tillage – this refers to the tillage operation following the primary tillage. It is done to create specific soil surface conditions before establishing the crop, at depths of <insert depth>. This is done using implements such as harrow, cultivators, packers, and hoes.
  - c. Combined Tillage – this refers to the tillage operation in which the primary and secondary tillage operations are done simultaneously using two or more implements to simplify, control, and reduce the number of trips in the field.
  - d. Broadcast Tillage – an entire area is tilled.
  - e. Partial Tillage – only bands or strips of land in an area is tilled.
  - f. Oriented Tillage – operations are done based on specific paths and directions i.e. sun, prevailing winds, field base lines, etc.
  - g. Rotary Tillage – Tillage is done using power-driven rotary action.
3. Specific Tillage Operations
    - a. Anchoring – tillage to partially bury plant residues and artificial mulches.
    - b. Bedding / Ridging / Listing – tillage operation to form a ridge or a furrow.
    - c. Bulldozing – pushing or rolling of soil by a steeply inclined blade.
    - d. Harrowing – usually used interchangeably with secondary tillage, it is an operation to pulverize and smoothen the soil in preparation for planting.
    - e. Incorporating – mixing and dispersing foreign material into the soil.
    - f. Middlebreaking – the formation of a furrow midway between two previous rows of plant.
    - g. Offbarring – cutting and throwing of the soil away from the base of the plant.
    - h. Moldboard Plowing – a type of primary tillage operation where the soil is cut and then partially or completely inverted.
    - i. Chisel Plowing – a type of primary tillage where the soil is mechanically loosened without inversion.
    - j. Subsoiling – a type of primary tillage at depths below 350 mm.
    - k. Vertical Mulching – a form of soil manipulation where vertical bands of mulching material is inserted into a slit in the soil.
  4. Tillage Patterns
    - a. Circuitous Pattern
    - b. Headland Pattern
    - c. Land System / Gathering Pattern

## VII. Crop Establishment

### A. Sexual Propagation

The most efficient and widely use method of propagation for most cultivated crops. New plants are produced through seeds which are produced through the union of male and female gametes.

#### 1. Types of Flowers

- a. Staminate – male flowers; having only the stamen.
- b. Pistillate – Female flowers; having only the pistil.
- c. Hermaphroditic – perfect flower; having both stamen and pistil.
- d. Monoecious – staminate and pistillate flowers grow on the same plant.
- e. Dioecious – staminate and pistillate flowers grow on different plant.
- f. Andromonoecious – staminate and hermaphroditic flowers grow on the same plant.

#### 2. Breeding Method for Seed Production

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- a. Open-Pollination – the natural process of pollination where plants are pollinated by natural forces such as wind, insects, birds, etc.
- b. Self-Pollination – type of pollination where the male and female gametes come from the same plant.
- c. Cross-Pollination – type of pollination where the male and female gametes come from different plant.
- d. Hybridization – the deliberate and controlled cross pollination of two different varieties in pursuit of producing a superior variety.

### 3. Seed Classes

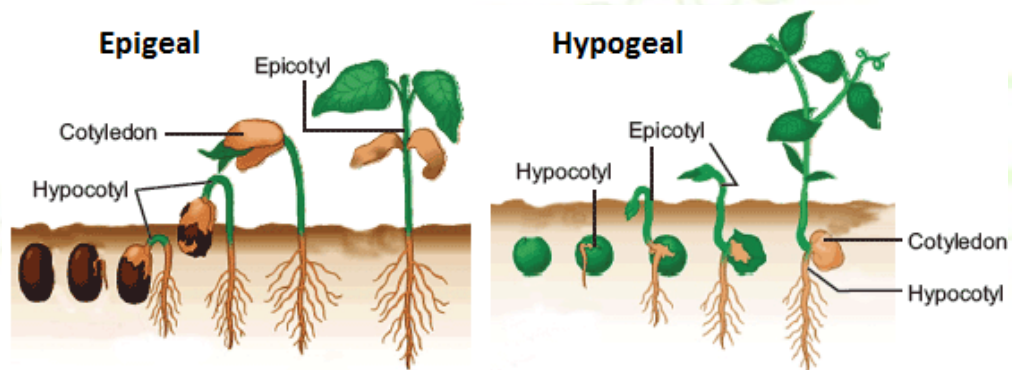
- a. Breeder Seeds
- b. Foundation Seeds
- c. Registered Seeds
- d. Certified Seeds

### 4. Seed Performance

- a. Seed Vigor
- b. Seed Enhancement
- c. Seed Priming

### 5. Types of Seed Germination

- a. Epigeal Germination – the hypocotyl elongates and pushes the cotyledons above the ground. Then, the epicotyl rapidly elongates, lifting the cotyledons.
- b. Hypogeal Germination – the epicotyl emerges from the ground, leaving the cotyledons below the ground surface.



### 6. Seed Storage Behaviors

- a. Seed Dormancy – a viable seed prevents germination even when subjected under favorable conditions which may be exogenous, endogenous, or imposed.
- b. Seed Quiescence – a viable seed cannot germinate due to unfavorable conditions.

### 7. Hastening Seed Germination

- a. Scarification – the process of removing or altering the seed coat to make the seed more permeable to water and air. This may be done physically (soaking in water), mechanically (filing, piercing, or grinding), and chemically (using sulfuric acid and other solvents).
- b. Stratification – the process of softening the seed coat by placing the seeds between layers of moist sand, soil, or sawdust at high or low temperatures.
- c. Vernalization – subjecting the seeds to low temperature prior to germination.

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- d. Embryo Culture – the process of removing the embryo from the seed and placing it in a sterilized culture to germinate.

## 8. Planting Methods

- a. Direct Seeding - seeds are directly sown into the soil to germinate and grow. It is the most economical way of sexual propagation but may not have the most efficient seed germination.
- Broadcasting - spreading / scattering of seed on top of the soil / seed bed either manually or mechanically. It is a quick and inexpensive method but may have an uneven distribution of seeds.
  - Drilling - Seeds are sown in drilled holes and the covered. This is done using a seed driller. Fertilizers may be incorporated during seeding.
  - Dibbling - involves placing seeds in small holes and then covered with soil using a tool called dibbler. This is usually applied on hilly lands.
  - Hill Dropping
  - Check Row Planting
- b. Transplanting – this involves raising the seeds in a nursery until it is viable to be planted in the field. This is a time-consuming process but ensures the viability of the seedling.

### Methods of Raising Rice Seedlings

- a. Wet-Bed Method - Seeds are sown in raised beds with continuous irrigation until it is ready for transplanting in 15-21 days.
- b. Dry-bed Method - Seeds are sown in raised beds in dry soil condition.
- c. Dapog Method - pre-germinated seeds are sown in a flat cemented or puddled soil with reliable irrigation and is ready to be transplanted after 10-14 days. An advantage of dapog method is that the uprooting of the seedlings is easier compared to bed methods.

## 9. Seeding Rate and Plant Density Computation

- a. Plant Density – refers to the number of plants per unit area.

$$\text{Plant Density} = \frac{A}{S_{\text{plant}} \times S_{\text{row}}}$$

Where A = Area

$S_{\text{plant}}$  = plant spacing

$S_{\text{row}}$  = row spacing

- b. Seeding Rate – refers to the amount of seeds needed which may be expressed as no. of seeds or weight of seeds.

$$\text{Seeding Rate} = \frac{PD \times \text{No. of Seeds per hill}}{\text{germination rate} \times \text{purity rate}}$$

Where PD = plant density

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## B. Asexual Propagation

The process of propagating crops using vegetative parts which is possible due to the regenerative properties of some plant organs

### 1. Types of Asexual Propagation

- a. Apomixis – the process of producing viable seeds without pollination, basically skipping fertilization. A parent plant develops seeds mitotically, which in effect produces genetically identical offsprings.
- b. Separation and Division – the process of separating naturally detachable organs or removing and cutting specialized vegetative structures from the mother plant. The following are the specialized plant structures used for separation and division:

|  |                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Bulbs specialized underground organ characterized by layers of fleshy leaves enclosing the flower primordium.</b>          |
|  | Bulbil – aerial plantlets which develop on the axils, leaves, or stalks of the parent plant.                                  |
|  | Corm – a modified swollen underground base of a stem enclosed by dry scale leaves.                                            |
|  | Crown – the base or pan of a plant at the ground surface from which new shoots are produced.                                  |
|  | Rhizomes – a specialized structure in which the main axis or stem of the plant grows horizontally beneath the ground surface. |
|  | Runners                                                                                                                       |
|  | Slips                                                                                                                         |
|  | Stolons                                                                                                                       |
|  | Suckers                                                                                                                       |
|  | Tubers                                                                                                                        |
|  | Tuberous Roots                                                                                                                |

- c. Cutting – a portion of the plant—such as a stem, leaf, or root—is cut and placed under favorable condition to induce the formation of roots and shoots.
- d. Layering – the process of growing adventitious roots on a stem while it is still attached to the parent plant.

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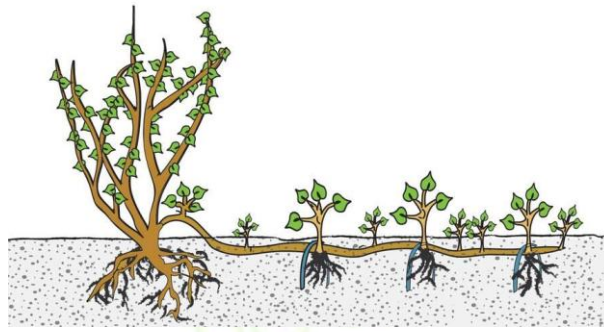


Figure 1. Illustration of compound layering.

- e. Grafting – the process of connecting two plants together for the purpose of uniting and growing them together as one plant.

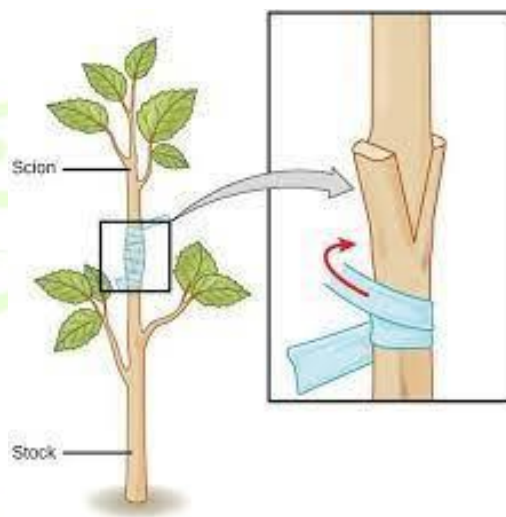


Figure 2. Process of grafting.

- f. Budding – the process of introducing the bud of a plant to grow on the branch of another plant.

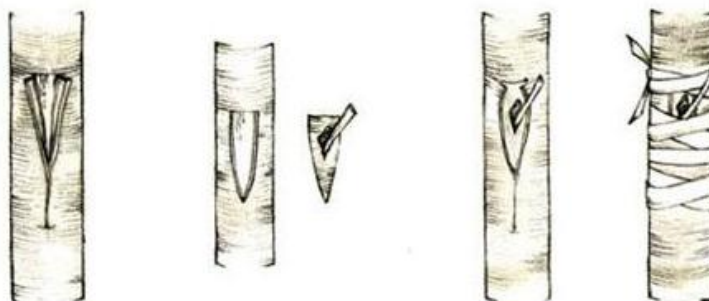


Figure 3. Illustration of budding.

- g. Inarching – the process of uniting two plants while still growing on their roots.

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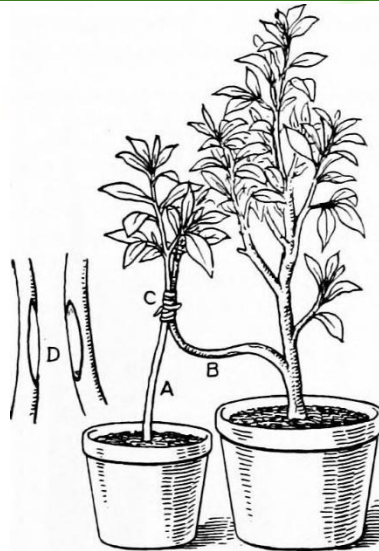


Figure 4. Illustration of inarching.

h. Tissue Culture – the process of growing plants from the cells of a parent plant.

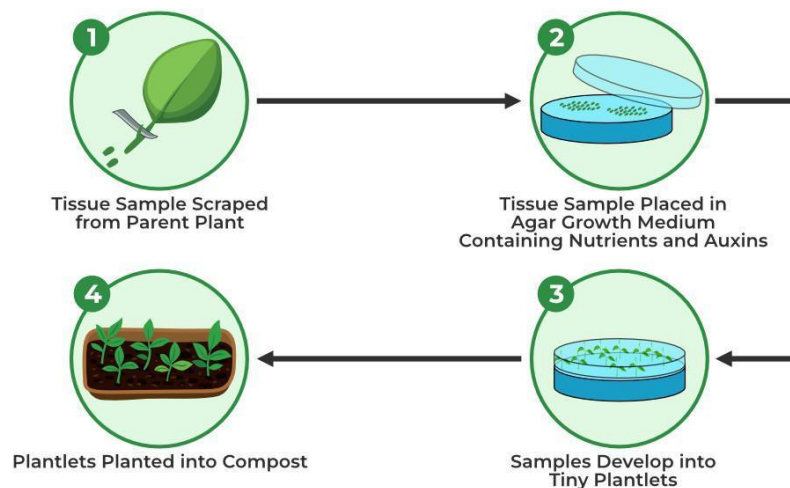


Figure 5. Process of tissue culture.

## C. Growth Stages of Crops

The length of time needed to grow varies from crop to crop. The general stages of crop growth are the following:

1. Germination - it is the process where a seed awakens from dormancy, sprouting its first roots and shoots. The seed absorbs water and nutrients which activates enzymes inducing the emergence of the radicle.
2. Seedling Phase - this is a time of delicate, rapid growth. The first true leaves emerge, and this is when the plant starts to develop its ability to photosynthesize and sustain itself.
3. Vegetative Growth - the plant focuses on growing strong. Leaves stretch out to soak up as much sunlight as possible, and roots push deeper into the soil to anchor the plant and absorb water and nutrients.

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4. Reproductive Growth - the plant focuses on reproduction—growing flowers, seeds, and fruits.
5. Maturity and Senescence

## VIII. Irrigation

Selecting the proper method of irrigation to be applied is important to ensure efficient and effective utilization of irrigation resources. The following are the factors that should be considered during selection:

1. **Crop Requirements** – irrigation is crop-dependent. An agricultural engineer should assess the crop's physiology, water needs, and growth stage to ensure that appropriate irrigation methods will be applied.
2. **Edaphic Factors** – the property of the soil predetermines the type of crops that may be planted and the type of irrigation that can be used; hence, it is a crucial factor in crop production.
3. **Climatological Factors** – the macro and microclimate in the area must be assessed. For example, sprinkler irrigation should not be used in areas that are relatively windy.
4. **Topography** – the irrigation method should account for the slope of the land. For example, drip irrigation is used for locations with uneven terrains because it is adaptable to variable elevation.
5. **Water Availability** – the availability of water resources should be taken into consideration to ensure that the water needs of the crop will be satisfied.
6. **Economic, Environmental, and Social Considerations**

### A. Surface Irrigation Methods

1. **Basin Irrigation** - a type of surface irrigation where areas of land surrounded by dikes (referred to as basins) are flooded with irrigation water, such as in rice production. Applicable to finely textured soils with low permeability.



Figure 15. Basin irrigation in lowland rice production.

2. **Border Strip Irrigation** – a type of surface irrigation where water is allowed to run down the slope of the strip of the land, guided by two adjacent borders. Water is allowed to flow in shallow sheets allowing it to enter the soil as it advances. This method is applicable to soils with medium infiltration rates.

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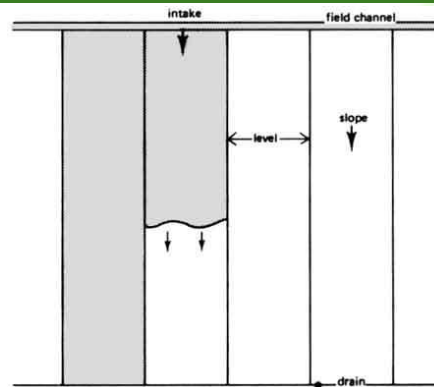


Figure 16. Border Strip Irrigation.

3. **Furrow Irrigation** – a type of surface irrigation where water is delivered into the field through small parallel channels or furrows in between rows of crops. Corrugation irrigation, a sub-variant of furrow irrigation, refers to the application of water into small rills in the soil.

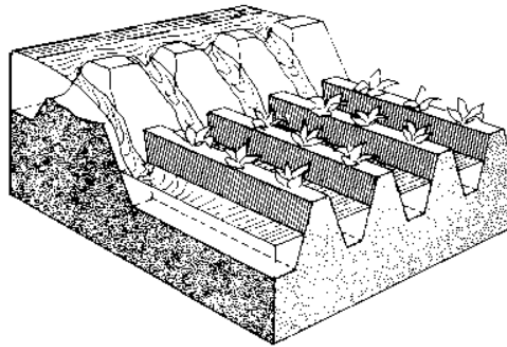


Figure 17. Furrow Irrigation.

4. **Wild Flooding / Uncontrolled Flooding** – a type of surface irrigation where water is allowed to flow freely into the field without guides. This method is relatively cheaper than the other methods; however, it is often inefficient due to uneven distribution of irrigation.

## B. Overhead Irrigation Methods

1. **Sprinkler Irrigation** – a type of overhead irrigation suited for most row, field, and tree crops where water is applied in the form of spray, mimicking natural rainfall.

### Types of Sprinkler Irrigation Systems

- a. **Set Systems** – sprinkler systems are operated in a fixed position.
  1. **Periodic Move** – laterals are moved manually through a series of positions during irrigation.
  2. **Fixed Systems** – sprinklers are fixed in the field and is not required to be moved during irrigation.
- b. **Continuous Move Systems** – sprinklers are operated while moving.
  1. **Traveling Gun Sprinklers** – a high-capacity sprinkler system mounted on a self-powered chassis and travels along a straight line while delivering irrigation.

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2. **Center-Pivot Systems** – irrigation is applied using a lateral pipe that moves in a circular pattern. One end of the lateral pipe is stationary, known as the pivot point, where it is connected to the water source.

3. **Linear Move Laterals** – combines the structural design and guidance system of a center-pivot with a moving water supply system, similar to that used in traveling sprinklers, allowing it to move in a straight line while irrigating

c. **Other Types**

1. Perforated Pipes
2. Hose-Fed Sprinklers

2. **Drip Irrigation** – also known as trickle irrigation, this refers to the method of irrigation where water is delivered into the soil using emitters at a very slow rate. In drip irrigation, water is applied closely to the plants, wetting only the part in which the roots grow, making it applicable even with limited water supply.

## C. Subsurface Irrigation Methods

1. **Buried Perforated Pipes** – irrigation is delivered through perforated pipes buried within the soil allowing water to be applied directly in the rootzone.
2. **Water Table Management Method** – water is pumped slowly and continuously into an open ditch to maintain a near constant water table.

## IX. Fertilizer Application

### A. Fertilizer Component

1. **Nutrient Component**
  - Macronutrients
  - Micronutrients
2. **Carrier Component**

### B. Methods of Fertilizer Application

- a. **Broadcast Method** – fertilizer is applied uniformly over the entire field.
  1. **Basal Method** – fertilizer is broadcasted over the entire area followed by cultivation to incorporate the fertilizer in the soil. This is done before planting.
  2. **Top Dressing** - Fertilizer is applied overhead on already growing seedling followed by overhead watering.
- b. **Localized Method** – fertilizer is applied on a specific area near the plants.
  1. **Banding Method** - fertilizer is applied in bands or strips on one or both sides of the seed or plant.
    - Row Method
    - Hole Method
- c. **Side Dressing**
- d. **Foliar Method**
- e. **Tank Method (Fertigation)**

### C. Fertilizer Computation

## X. Pest and Weed Control

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## PRINCIPLES OF ANIMAL SCIENCE

### I. Anatomy and Physiology of Farm Animals

Animals have several different anatomical systems, each with specialized structures and functions that work together to maintain life. Here's an overview of the major systems and their roles:

#### A. Integumentary System

- Composed of the skin, hair, feathers, scales, claws, hooves, etc.
- This system helps protect the animal against injury, infection, and dehydration. It also helps the body to regulate temperature.

#### B. Skeletal System

- Composed of bones, cartilage, ligaments, tendons.
- Provides structural support and shape.
- Protects internal organs (e.g., skull protects the brain).
- Facilitates movement (works with muscular system).
- Stores minerals (calcium, phosphorus) and produces blood cells (bone marrow).

#### C. Muscular System

- Composed of skeletal muscles, smooth muscles, cardiac muscles.
- Enables movement (locomotion, posture).
- Supports circulation (heart muscles pump blood).
- Aids in digestion (smooth muscles in the gut).

#### D. Nervous System

- Composed of the brain, spinal cord, nerves, sensory organs.
- Processes sensory information (sight, sound, touch).
- Coordinates voluntary and involuntary actions. Regulates other systems (homeostasis).

#### E. Endocrine System

- Composed of glands (pituitary, thyroid, adrenal, pancreas, etc.) and hormones.
- Regulates growth, metabolism, and reproduction.
- Maintains homeostasis (e.g., insulin regulates blood sugar).

#### F. Cardiovascular System

- Composed of the heart, blood vessels (arteries, veins, capillaries), and blood.
- Transports oxygen, nutrients, hormones, and waste.
- Helps in immune response and temperature regulation.

#### G. Respiratory System

- Composed of lungs, gills, trachea, diaphragm (in mammals).
- Facilitates gas exchange (oxygen in, carbon dioxide out).
- Supports cellular respiration.

#### H. Excretory System

- Composed of the kidneys, bladder, ureters, urethra.
- Filters blood and removes waste (urea, excess salts).
- Maintains water and electrolyte balance.

#### I. Lymphatic System

- Composed of lymph nodes, spleen, white blood cells, antibodies.
- Defends against pathogens (bacteria, viruses).

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- Removes toxins and waste from tissues.

## J. Reproductive System

- Composed of the ovaries, testes, uterus, sperm ducts, etc.
- Produces gametes (sperm and eggs).
- Facilitates fertilization and development of offspring.

## K. Digestive System

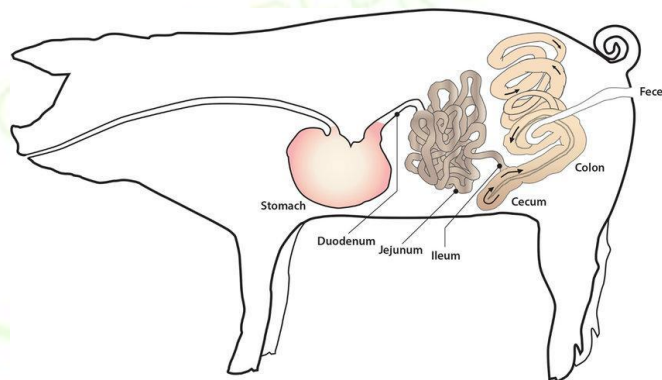
- Composed of the mouth, esophagus, stomach, intestines, liver, pancreas.
- Breaks down food into nutrients.
- Absorbs nutrients for energy and growth.
- Eliminates waste (feces).

### a. Important Terminologies in Animal Digestion

1. Prehension - the act of conveying the food into the mouth.
2. Mastication - the act of chewing.
3. Insalivation - the mixing of feed with saliva.
4. Deglutition - the act of swallowing.
5. Rumination - the process of regurgitating food bolus from the rumen, chewing it again, and then swallowing for the purpose of improving digestion.
6. Digestion - breaking down of food particles into suitable products for absorption
7. **Absorption** - the process of absorbing the nutrients from the feed into the blood to be supplied to the rest of the body.

### b. Types of Animal Digestive System

1. **Monogastric Stomach** - The stomach only has one compartment such as in swine and horses. The digestive pathway starts with the mouth - esophagus - stomach - small intestine - large intestine - rectum - anus.

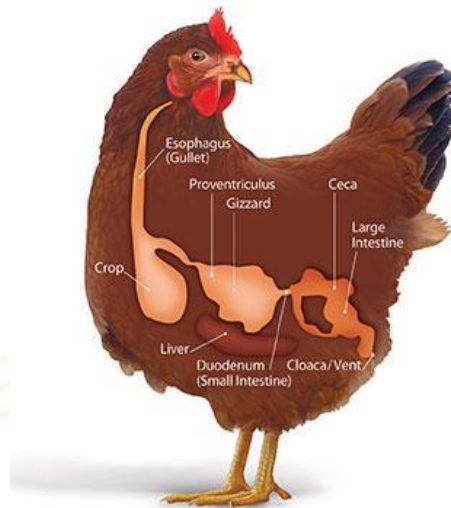


2. **Modified Simple Stomach** - The stomach system of poultry animals. The digestive pathway starts with the mouth - esophagus - crop - proventriculus - gizzard - small intestine - large intestine - cloaca - vent.

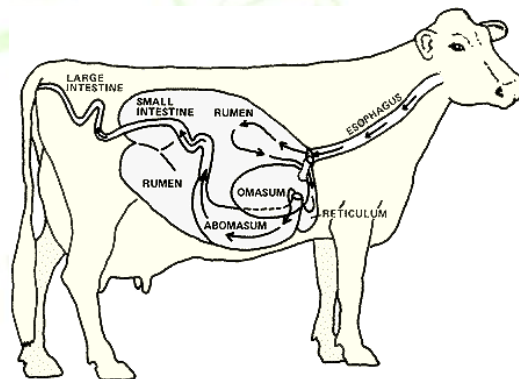
- a. **Crop** – serves as temporary storage for the moistening of feed.
- b. **Proventriculus** – considered as the true stomach of poultry animals where digestive juices break down feed.
- c. **Gizzard** – provides the mechanical breakdown of food.

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3. **Compound Stomach** - The stomach system of ruminants composed of four compartments. The digestive pathways starts with the mouth - esophagus - reticulum - rumen - (rumination) - reticulum - rumen - omasum - abomasum - small intestine - large intestine - rectum - anus.



- a. **Reticulum** - honeycomb lining; responsible for moving the feed from the rumen to the omasum and for the collection of foreign materials; aids in regurgitation.
- b. **Rumen** - contains papillae; the main site for the microbial fermentation of the feed to break down fiber.
- c. **Omasum** - called "many piles" or the "butchers bible" due to muscular laminae. Its function is to further grind (mechanical breakdown) before entering abomasum.
- d. **Abomasum** - the true stomach of the ruminants.



RETICULUM



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|                                                                                             |       |
|---------------------------------------------------------------------------------------------|-------|
|                                                                                             | RUMEN |
| <br>OMASUM |       |

## II. Animal Nutrition

- Nutrition refers to the breaking down food through digestion and assimilating nutrients through absorption for the purpose of supporting life.
- Nutrients refers to any essential substance found in feed that support life, growth and productivity.

### A. Nutrient Classes and Their Functions

1. **Water** - the cheapest and most abundant nutrient. It is essential for transporting other compounds, for regulating the body temperature, and other physiological processes.
2. **Carbohydrates** - functions as the main energy source of animals available in the forms of sugars, starches, and fibers. It is stored in the animal body in form of fats.
3. **Fats** - provides energy and aids in the absorption of fat-soluble vitamins.
4. **Vitamins** - acts as a catalyst for biochemical processes such as metabolic activities. It cannot be synthesized by the animal body.
5. **Minerals** - aids in the development of the structural portion of the animal body.
6. **Proteins** - act as the building blocks for muscle development, enzyme production, and hormone regulation. This is a key component in animal feed formulation.

### B. Sources of Animal Nutrition

1. **Roughages**
2. **Supplemental Feed**
  - a. **Feed Formulation**

### C. Evaluation

1. **Average Daily Gain**
2. **Feed Efficiency**

## III. Animal Reproduction

### A. Methods of Animal Reproduction

1. **Sexual Reproduction** – animals produce offsprings through mating.
2. **Asexual Reproduction** – animals reproduce without mating such as artificial insemination and embryo transfer.

### B. Methods of Fertilization

1. **Internal Fertilization** - the male gamete is deposited inside the female where it will come into contact with the female gamete.
  - a. **Oviparity** - the embryo is formed inside the body and deposited as an egg.
  - b. **Viviparity** - the embryo is formed inside the body, obtains the nutrition of the blood of the parent, and then emerges fully developed from the female's body.

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c. **Ovoviviparity** - The embryo is formed inside the body, obtains nutrition from the egg yolk, hatches, and released from the body.

2. **External Fertilization** - parental animals expel their gametes into another medium such as water without necessarily coming into contact with each other.

## C. The Estrous Cycle

1. **Proestrus** – described by the regression of the corpus luteum, estrogen production is increasing.
  2. **Estrus** - period of sexual receptivity due to high levels of estrogen. Physical manifestations of estrus includes reddening and swelling of the vulva, mucus discharge in the vagina, frequent urination, restlessness and lack of appetite, mounting (female stands still). The best time to inseminate or to mate is at the end of estrus.
  3. **Metestrus Cycle** – characterized by the formation of the corpus luteum and the increasing production of progesterone.
  4. **Diestrus** – the corpus luteum is fully formed and is highly active in the production of progesterone.
- a. Classification of Animals based on Estrous Cycle
1. **Monoestrous** – becomes in heat once a year.
  2. **Polyestrous** – becomes in heat multiple times throughout the year.
  3. **Seasonally Polyestrous** – becomes in heat in certain seasons.

## D. Other Important Terms in Animal Reproduction

1. **Fertilization** – the union of the male and female gamete.
2. **Conception** – the fertilization and implantation of the embryo into the uterus of the female parent animal.
3. **Fecundation**
4. **Fecundity**
5. **Prolific**
6. **Parturition** – the act of giving birth to a live young.
7. **Oviposition** – the act of laying eggs.
8. **Dystocia** – refers to the difficulty in labor.
9. **Afterbirth** – the technical term for the placenta and allied membranes.
10. **Deadborn / Stillborn** – an animal that is dead at birth.
11. **Colostrum** – the first milk produced by the female animal.
12. **Agalactia** – failure to secrete milk.
13. **Mastitis** – inflammation of the mammary glands.
14. **Metritis** – inflammation of the uterus.
15. **Multiple Ovulation** – the process of producing multiple egg cells in a cycle.
16. **Oxytoxin**
17. **Estrogen**
18. **Progesterone**

E. **Animal Breeding** – refers to the art and science of genetic improvement of farm animals.

### a. Systems of Mating

1. **Random Mating** - system of mating where reproductively active animals are free to mate with other animals of the opposite sex, regardless of genotype and phenotype.
2. **Non-random Mating** - system of mating where animals are selectively reproduced and does not have an equal chance of mating with other animals of the opposite sex. The following are the forms of non-random mating:
  - **Assortative Mating** - Mating of animals which are more phenotypically similar.

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- Disassortative Mating - Mating of animals which are less phenotypically similar.
- b. Systems of Breeding
  1. Selection - only animals with desirable traits are allowed to reproduce to control the genetic characteristics of the offspring.
  2. Inbreeding - animals that are related by descent or familial relationship are allowed to mate to increase the frequency of a desirable trait. The methods of inbreeding are as follows:
    - Close Breeding - Animals that are closely related are allowed to mate such as (dam x offspring), (sire x offspring), (offspring x offspring).
    - Line Breeding - Animals that are related by ancestry, but not as closely related as in close breeding, are allowed to mate.
  3. Cross Breeding - animals of different breeds are allowed to mate to combine desirable traits, potentially resulting to heterosis.
- c. Other Important Terms in Animal Breeding
  1. Backcrossing
  2. Progeny
  3. Heterosis / Hybrid Vigor

## IV. Terminologies in Poultry Production

The term poultry generally refers to avians (birds) which render service to humans as food (meat and egg), entertainment, and others. They are primarily raised for meat production (broiler) and egg production (layer).

### A. Categories of Poultry Production

1. **Breeding Farm** – a system used for researching the best matches of parent stocks that will produce high quality of stocks for the poultry producers.
2. **Layer Farm** – a system of poultry farming mainly focused on the production of eggs. Standards for animal housing for layer production is available on PAES 403:2001.
3. **Broiler Farm** – a system of poultry farming mainly focused on meat production. Standards for animal housing for layer production is available on PAES 402:2001.

### B. Terminologies in Poultry Reproduction

1. **Incubation** – the process of applying heat to the eggs to induce the development of the embryo.
2. **Egg Candling** – the process of evaluating the fertility and viability of an egg by placing it against a bright light which then emphasizes the development of the embryo inside the egg.
3. **Brooding** – the process of applying heat to the young poultry animal for approximately four weeks until they no longer need supplemental heat.

#### Temperature Requirement for Brooding

|                 |   |                                |
|-----------------|---|--------------------------------|
| Week 0 – Week 1 | : | 32.2 – 35 degrees Celsius      |
| Week 1 – Week 2 | : | 29.4 – 32.2 degrees Celsius    |
| Week 2 – Week 3 | : | 26.7 – 29.4 degrees Celsius    |
| Week 3 – Week 4 | : | 26.7 degrees Celsius and below |

### C. Chicken (*Gallus gallus domesticus*, or simply *Gallusd gallus*)

- a. Classes based on Purpose

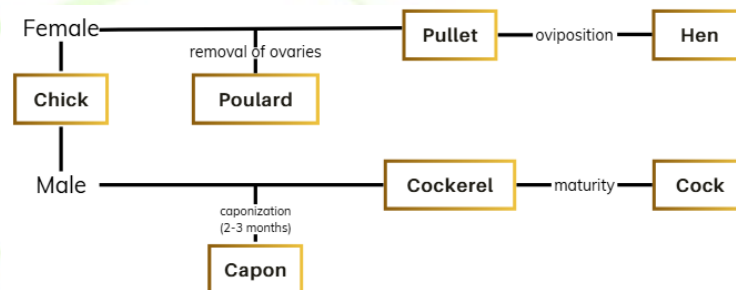
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1. Egg Class (Layer) - comparatively small, generally non-sitters (does not display brooding behavior), and lays large, white-shelled eggs. Common breeds are Leghorn, Minorca, Anconas, Mikawa
2. Meat Class (Broiler) - comparatively large and are generally poor egg layers. Common breeds are Brahmas, Cornish Cross, White Rocks, Cobb.
3. General Purpose Class - medium sized chickens which are potentially good layers and are fast-growing when young. Common breeds are Rhode Island Red, Plymouth Rock, New Hampshire.
4. Fighting Class - used for cock-fighting purposes such as Rubkle, Hulsey, Claret, and Oasis.
5. Fancy Class – primarily raised as ornamentals or pets such as Bantams and Frizzles.

## b. General Terminologies

1. Cock – mature male fowl.
2. Hen – mature female fowl.
3. Chick – juvenile fowl of any sex.
4. Straight-run Chicks – unclassified chicks.
5. Brood – a group of chicks.
6. Caponization – the process of removing the testicles of male fowls to improve meat production.
7. Capon – fattening male fowls whose testicles had been removed.
8. Cockerel – young male chick below one year of age.
9. Pullet – young female chick below one year of age that has not laid eggs yet.
10. Poulard – pullet whose ovaries had been removed.



## D. Duck (*Mallard (Anas platyrhynchos)* and Muscovy – (*Cairina moschata*))

1. Drake – term for male breeding duck.
2. Duck hen – term for female breeding duck.
3. Duckling – term for the offspring of ducks.

## E. Quails (*Coturnix coturnix*)

## F. Turkey (*Melleagris gallopavo*)

1. Tom or gobbler – term for male breeding turkey.
2. Turkey hen – term for female breeding turkey.
3. Poult - a young, unclassified turkey.

## G. Goose (*Anser domesticus*)

1. Gander – term for male breeding goose
2. Goose – term for female breeding goose
3. Gosling – term for young goose / offspring.

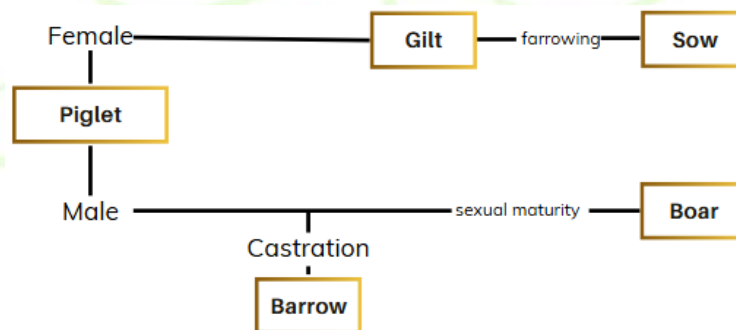
## V. Terminologies in Swine Production

# FUNDAMENTALS OF CROP AND ANIMAL SCIENCES

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ABELE 2024

## A. General Terminologies in Swine

1. Swine - umbrella term for animals under the Suidae family.
2. Hog – term for male swine above one year old.
3. Pig – term for young swine below one year old.
4. Boar – term for mature male breeding swine.
5. Sow – term for female breeding swine that has given birth.
6. Gilt – term for female breeding swine below one year old that has not given birth yet.
7. Shote – young swine of any age weighing below 50 kg.
8. Barrow – term for male swine castrated before the development of the secondary sexual characteristics.
9. Boar Taint – an undesirable odor and taste of the meat derived from uncastrated swine.
10. Farrowing – term referring to the act of giving birth in swine.
11. Litter – a collective term referring to all the piglets born in one farrowing.
12. Piglet – term for the offspring of swine.
13. Pork – meat derived from swine.
14. Weaning – the process of separating the young from the dam and transitioning them into a solid-based diet.
15. Weaner – a young that has been separated from the dam and is no longer reliant on milk for nourishment.



4. Note: Standards for animal housing for swine production is available on PAES 401:2001.

## B. Common Swine Breeds

|                                                                                                                       |                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Landrace</b><br>                | First developed in Denmark. It is generally white in color (may have freckles or spots) and is known as the longest breed of swine. Prolific in the local swine raising environment. |
| <b>Large White / Yorkshire</b><br> | Developed in England, this breed is entirely white in color. Known for raising large litters and are good milk producers.                                                            |
| <b>Duroc</b>                                                                                                          | Originated in the Eastern US. They have solid colors ranging from very light golden to very                                                                                          |

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|                                                                                                               |                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | dark red (mahogany). Considered as a superior breed due to its growth rate, feed efficiency, and resistance to stress.                                                                                |
| <b>Pietrain</b><br>          | Originated in Belgium. Often called "muscle pigs" due to its muscle development in the ham, loin, and shoulder areas. Generally slow growers and are highly susceptible to stress.                    |
| <b>Hampshire</b><br>         | Originated in Southern England. They are black in color with a white belt around the shoulder, body, and foreleg.                                                                                     |
| <b>Berkshire</b><br>        | Originated in South Central England. They are known for their high-quality meat. They have a short and upturned nose and is black in color with white points in the feet, forehead, and tail.         |
| <b>Poland of China</b><br> | Originated in Ohio. They are known for their large size. They are black in color with white marking on their nose, feet and tail. Differentiated from the Berkshire breed due to their drooping ears. |

## VI. Ruminants

Ruminants are animals whose stomachs are divided into four compartments for the efficient digestion of cellulose fiber from roughages. The following are example of ruminants:

### A. Cattle (Continental Breed (*Bos taurus*) or Oriental Breed (*Bos indicus*))

The term cattle refer to animal farms from the Bovidae family and the genus Bos.

1. Bull – term for male breeding cattle of any age.
2. Cow – term for mature female cattle which had already given birth.
3. Heifer – term for young female cattle less than three years of age which has not given birth.
4. Steer – male cattle castrated before the development of secondary sexual characteristics.
5. Stag – male cattle castrated after sexual maturity.
6. Beef – meat derived from mature cattle.
7. Veal – meat derived from a calf.
8. Calving - term for the act of giving birth in cattle

# FUNDAMENTALS OF CROP AND ANIMAL SCIENCES

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- a. Production Systems
  1. Ranching or Extensive System
  2. Feedlots
  3. Backyard Cattle Raising

## B. Carabao (*Bubablus bubalis*)

1. Carabull - technical term for male breeding carabao
2. Caracow - technical term for female breeding carabao
3. Caraheifer - technical term for young female carabao less than three years of age which has not given birth.
4. Carasteer - term for male carabao which was castrated before the development of the secondary sexual characteristics.
5. Carastag - term for male carabao which was castrated after sexual maturity
6. Carabeef - meat derived from carabaos.

## C. Goat (*Capra hircus*)

1. Buck / Billy - term for male breeding goat.
2. Doe - term for female breeding goat.
3. Doe-in-kid - pregnant doe.
4. Kid - term for the offspring of goats, regardless of sex.
5. Wether - term for male goat which was castrated before the development of secondary sexual characteristics.
6. Stag - term for male goat which was castrated after sexual maturity.
7. Kidding - term for the act of giving birth in goats.
8. Chevron - term for the meat derived from goats.

## D. Sheep (*Ovis aries*)

1. Ovine - umbrella term for sheeps.
2. Sheep - ovine above one year of age, regardless of sex
3. Lamb - ovine below one year of age, regardless of sex.
4. Ram - term for mature male breeding sheep.
5. Ewe - term for mature female breeding sheep.
6. Wether - term for male sheep which had been castrated before the development of secondary sexual characteristics.
7. Stag - term for male sheep which had been castrated after sexual maturity.
8. Shearer - sheep primarily raised for wool production.

Fleece - refers to the wool covering of the sheep

Wool - refers to the textile fiber derived from sheep

Pelt - refers to the wool and skin of the sheep.

Carding - refers to the process of